Milind Nakul

PhD Student, Georgia Institute of Technology

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Research Interests

My research interests lie at the intersection of high-dimensional statistics, optimization, and reinforcement learning. I am particularly passionate about developing theoretically motivated algorithms to address fundamental challenges in these fields. Recently, my work has focused on creating techniques that enable effective learning from dependent data.

Education

Present – Aug 2023 | **Georgia Institute of Technology** | **Atlanta, GA, US**
Ph.D., Machine Learning (H. Milton Stewart School of Industrial & Systems Engineering)
Minor: Mathematics | GPA: 3.85/4.0
Advisors: Prof. Ashwin Pananjady, Prof. Vidya Muthukumar

Apr 2023 – Jul 2018 | **Indian Institute of Technology Kanpur** | **Kanpur, India**
B.Tech. and M.Tech. (Dual Degree), Electrical Engineering
Minor: Machine Learning & Applications | CPI: 9.6/10
Thesis Title: Momentum-Based Variance Reduced Algorithms for Imitation Learning (Link)
Advisor: Prof. Ketan Rajawat

Experience

Aug 2026 – May 2026 | **Dolby Laboratories** | **San Francisco, CA, US**
PhD Research Intern | Spend Compute Where It Matters: Input-Aware Sparsity Routing for Large Models
Built an input-aware routing framework that trains a lightweight router to dynamically select a per-input sparsity level. Generalized the approach across vision and language, cutting active-model memory by 33–43% on Qwen2.5-7B while matching dense ResNet-152/ImageNet top-1 accuracy (79.5%) at 31% average sparsity.

Present – Jan 2024 | **Georgia Institute of Technology** | **Atlanta, GA, US**
Graduate Research Assistant | Advisors: Prof. Ashwin Pananjady, Prof. Vidya Muthukumar
High-dimensional statistics and reinforcement learning. Developed a multi-scale replay algorithm that solves the policy evaluation problem in reinforcement learning without mixing information, demonstrating how "experience replay" can be applied in a statistically efficient manner with dependent data.
Distribution estimation from dependent data (since Sept 2024). Proposed a new estimator, implementable in linear time, for the stationary distribution of dependent data, with applications ranging from ecology and natural language modeling to quantifying hallucinations in LLMs.

Apr 2023 – May 2022 | **IIT Kanpur | Signal Processing in Networks (SPiN) Lab** | **Kanpur, India**
Graduate Research Assistant | Advisor: Prof. Ketan Rajawat
Worked on stochastic gradient-based algorithms for imitation learning. Implemented DAGGER, SMILe, and forward training on several OpenAI Gym environments, and developed an SGD-based imitation learning algorithm that reduces the number of calls made to the expert.

Oct 2022 – Jan 2022 | **IIT Kanpur | Intelligent Wireless Networks (IWiN) Lab** | **Kanpur, India**
Undergraduate Researcher | Advisor: Prof. Rohit Budhiraja
Worked on variational inference for channel estimation in reconfigurable intelligent surface (RIS) assisted mmWave systems, developing a variational expectation-maximization algorithm with improved time complexity.

Jul 2021 – May 2021 | **Cisco Systems** | **Bangalore, India**
Software Engineering Intern | Manager: Rajavel Ganesamoorthy
Developed a custom tool to parse command outputs and extract field descriptions from the IP multicast routing table using regular expression matching, and integrated the project into the Automation Framework platform used at Cisco as a unit test automation template.

Publications/Patents

P=In Preparation, S=In Submission, C=Conference, J=Journal

[P.1] **Spend Compute Where It Matters: Input-Aware Sparsity Routing for Large Models**
Milind Nakul, Kadierdan Kaheman, Jason Cloud
Patent Pending, United States Patent and Trademark Office; Invention Disclosure Form filed with Dolby Laboratories **[2026]**

[P.2] **Next Token Functional Estimation**
Milind Nakul, Vidya Muthukumar, Ashwin Pananjady
Under preparation **[2026]**

[S.2] **Multiscale Replay: A Robust Algorithm for Stochastic Variational Inequalities with a Markovian Buffer** [🔗] [arXiv:2601.01502]
Milind Nakul, Tianjiao Li, Ashwin Pananjady
Under review at Mathematics of Operations Research **[Math. OR]**

[C.1] **Estimating Stationary Mass, Frequency by Frequency** [🔗] [arXiv:2503.12808]
Milind Nakul, Vidya Muthukumar, Ashwin Pananjady
Conference on Learning Theory (COLT); Under review at SIAM Journal on Mathematics of Data Science **[COLT'25, SIMODS]**

[J.1] **Variational Learning Algorithms for Channel Estimation in RIS-assisted mmWave Systems** [🔗] [arXiv:2211.14495]
Milind Nakul, Anupama Rajoriya, Rohit Budhiraja
IEEE Transactions on Communications **[IEEE TCOM'23]**

Talks and Presentations

"Multiscale Replay: A Robust Algorithm for Stochastic Variational Inequalities with a Markovian Buffer", INFORMS Optimization Society Conference (Talk), Atlanta, GA **Mar'26**

"Estimating Stationary Mass, Frequency by Frequency", Conference on Learning Theory (COLT) (Talk), France **Jul'25**

"Estimating Stationary Mass, Frequency by Frequency", International Statistics Conference (IISA) Conference (Poster), Lincoln, NE **Jun'25**

Honours and Awards

John Morris Fellowship, awarded by the H. Milton Stewart School of Industrial & Systems Engineering, Georgia Tech. 2023–24

Dr. Prateek Mishra Memorial Gold Medal, awarded by IIT Kanpur. 2023

Academic Excellence Award (equivalent to Dean's List), awarded by IIT Kanpur for exceptional academic performance. 2018–22

All India Rank 773 in JEE Advanced 2018, among 0.2 million applicants. 2018

All India Rank 634 in JEE Mains 2018, among 1.2 million applicants. 2018

Rank 13 in the State Level National Talent Search Examination (NTSE) organised by NCERT. 2015–16

Academic Service

Conference Reviewer	ALT 2027, NeurIPS 2026, ICML 2026, ALT 2026, COLT 2026
Journal Reviewer	Annals of Statistics
Webmaster	Stochastic Networks, Applied Probability, and Performance (SNAPP) Seminar

Teaching and Volunteering Roles

ISYE3133 Engineering Optimization, Georgia Tech — Tutor Aug'23 – Apr'24

Conducted weekly tutorial sessions, and mentored students on various topics in optimization.

EE675 Introduction to Reinforcement Learning, IIT Kanpur — Teaching Assistant Jan'23 – Apr'23

Conducted weekly tutorial sessions, and mentored students on various projects on applications of reinforcement learning.

EE381 Electrical Engineering Laboratory, IIT Kanpur — Teaching Assistant Aug'22 – Apr'23

Conducted lab and weekly tutorial sessions, and planned course logistics, teaching assessment, and evaluation for 150+ students.

Electronics Club, IIT Kanpur — Secretary May'19 – Jul'20

Organised lectures and workshops on machine learning for electronics, mentored participants in intra-IITK competitions and projects, and worked on emulating quantum algorithms (Grover's Search, QFT) on FPGA.

Relevant Coursework

Computer Science	Probabilistic Machine Learning, Deep Reinforcement Learning, Data Structures & Algorithms, Introduction to Machine Learning, Fundamentals of Computing, Embedded and Cyber-Physical Systems
Mathematics	High-dimensional Statistics and Optimization, High Dimensional Probability, Probability and Statistics, Real Analysis & Multivariate Calculus, Linear Algebra and Differential Equations, Complex Analysis
Engineering	Convex Optimization, Machine Learning for Wireless Communications, Analysis of Random Signals